

Basic Networking Commands.

23/1/26

Aim:-

Study of various network commands used in Linux and windows.

1) arp -a:-

Displays the Address Resolution protocol (ARP) table, which contains the IP addresses and their corresponding MAC (physical) addresses stored in the system.

Output:-

Interface : 192.168.56.1 --- 0xd

Internet Address	Physical address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	Static
224.0.0.22	01-00-5e-00-00-1b	Static
224.0.0.251	01-00-5e-00-00-fb	Static

2) Hostname:-

Displays the hostname (computer name) of the ~~local~~ machine.

Output:-

LAPTOP-J8UH0T4B

3) ipconfig /all:-

Displays detailed information about the network adapter including IP address, subnet n

default gateway, DNS server, DHCP server,
MAC addresses etc.

Output:-

Windows IP configuration.

Host name : LAPTOP-18UHOT4B
Primary Dns suffix :
Node Type : Mixed
IP Routing enabled : No

4) nbtstat -a :-

Displays the NetBIOS name table of a
remote or local computer and help troubleshoot
NetBIOS name resolution problems.

Output:-

Ethernet 5:

Node IP Address : [192.168.56.1] Scope Id : []

Host not found.

Ethernet :

Node IP Address : [172.16.11.50] Scope Id : []

Host not found.

5) netstat -a

Displays active TCP/UDP connections, listening
ports, routing table, and network statistics

Output:-

Interface List:
18...14 16 9e 6d 97 ab ... Realtek PCIe GbE family Controller
13...0a 00 27 00 00 ad VirtualBox Host-Only Ethernet Adaptor
10...54 14 f3 7d c8 d3 Microsoft Wi-Fi Direct Virtual Adapter #3
22...54 14 f2 7d c8 d2 Intel(R) Wireless-AC 9560 160MHz

b) nslookup google.com:-

Queries the DNS server to obtain the IP address of a domain name.

Output:-

Server: Unknown

Address: 172.16.8.1

Non-authoritative answer:

Name: google.com

Addresses: 2404:b800:4001:805::200e

142.250.67.46

f) pathping google.com:-

Combines the features of ping and Tracert to trace the path to a destination and calculate packet loss at each hop

Output:-

Tracing route to google.com [142.250.67.46]

Over a maximum of 30 hops:

0 LAPTOP-J8UH0T4B [172.16.11.50]

1 172.16.8.1

2 137.97.163.129

3 * * *

8) ping google.com:-

Check whether another computer or website is reachable over the network using ICMP Echo Request and Echo Reply messages.

Output:-

Pinging google.com [142.250.67.46] with 32 bytes of data:

Reply from 142.250.67.46 : bytes = 32 time = 4ms
TTL = 118

Reply from 142.250.67.46 : bytes = 32 time = 4ms
TTL = 118

Reply from 142.250.67.46 : bytes = 32 time = 10ms
TTL = 118

Reply from 142.250.67.46 : bytes = 32 time = 7ms
TTL = 118

9) route:-

Displays or modifies the IP routing table used by the operating system

Output:-

Manipulates network routing tables.

ROUTE [-f] [-P] [-4|-6] command [destination]

[MASK netmask] [gateway] [METRIC metric]

-f clears the routing table of all gateway entries. If this is used in conjunction with one of the commands, the tables are cleared prior to running the command

Linux Commands

10) ip address show:-

Displays and manages IP addresses, routing and network interfaces.

Output:-

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536

qdisc noqueue state UNKNOWN group default

link /loopback 00:00:00:00:00:00 brd

00:00:00:00:00:00

Inet 127.0.0.1/8 scope host lo

valid-lft forever preferred-lft forever

11) ifconfig :-

Displays network interface details such as IP address, and network statistics.

Output:-

```
etho: flags = 4163 <UP, BROADCAST, RUNNING, MULTICAST> mtu 1500
```

```
inet 172.17.197.94 netmask 255.255.240.0 broadcast 172.17.205.255
```

```
inet6 fe80::215:5dff:fe82:dc86  
Prefixlen 64 scopeid 0x20 <link>
```

12) mttr google.com:-

Combines Ping and Traceroute to display the network path and performance statistics.

Output:-

LAPTOP-J8UH0T4B (172.17.197.94) -> google.com

Host

	Loss%	Sent	Last
1. LAPTOP-J8UH0T4B.mshome.net	0.0%	128	0.7
2. 192.168.1.1	0.0%	146	4.7
3. 100.84.0.1	0.0%	163	5.7
4. (waiting for reply)			

13) tcpdump -i eth0 :-

Captures and displays network packets passing through a network interface for analysis.

Output:-

tcpdump: verbose output suppressed, use -v or
-vv for full protocol decode

listening on eth0, link-type EN10MB (Ethernet),
capture size 262144 bytes

10:15:21.123456 IP 192.168.1.10.52134 >

8.8.8.8.53 : UDP, length 32

14) ping google.com:-

Sends ICMP Echo request packets to verify network connectivity and measure response time.

Output:-

PING google.com (142.251.43.206) 56 (84) bytes
64 bytes from 1cmaa-bb-in-f14.1e100.net
(142.251.43.206): icmp_seq=1 ttl=117

time = 5.76 ms

Result:-

Thus, the various windows and Linux networking commands were executed successfully.